Saaf Hawa v2: Rebuilding Ground Truth & Smog Forecasting

Track: Air Quality | **Event:** HackInsight '26 | **Zero-Cost Open Source Stack**

1. Executive Summary & Problem Statement

In March 2025, public ambient air quality monitoring in Pakistan suffered a severe blind spot when U.S. State Department / EPA embassy monitors in Lahore, Islamabad, Karachi, and Peshawar stopped transmitting data to AirNow. For district education officers, health officials, and school heads across Punjab, this eliminated the only public reference-grade source of truth for high-stakes decisions (school closures, morning assembly cancellations, mask mandates). **Saaf Hawa v2** reconstructs ground truth from community sensor networks (PAQI) calibrated against the pre-halt embassy record, and couples it with NASA FIRMS transboundary fires and ECMWF CAMS to provide 24h quantile forecasts with calibrated conformal uncertainty.

2. Multi-Publisher Data Stack & Volume

• **Four Independent Publishers:** OpenAQ (PAQI), AirNow DoS archive (source not on track list), ECMWF/Copernicus (Open-Meteo), and NASA FIRMS.
• **Data Volume:** 56,388 city-hourly PM2.5 rows, 133,440 atmospheric features in DuckDB, backed by 28,652 AirNow historical calibration rows and 60,078 community sensor rows.
• **Data Contract:** Timestamps in UTC, deterministic row IDs ({CITY}-{YYYYMMDDTHHZ}), full SHA-256 provenance in manifest.csv.
• **Dual Quality Gates:** Pandera schema validation, sentinel conversion (-999 -> NULL), stuck-sensor detection (6h), and peer corroboration during extreme smog spikes (500-1000 ug/m3).

3. Scientific Evaluation vs Declared Baselines (Like-for-Like Held-out Test Split, 24h Embargo)

Forecast Horizon	Baseline A (Persistence) MAE	Baseline B (Seasonal-Hour) MAE	Saaf Hawa LightGBM+CQR MAE	80% Conformal Coverage
+6 Hours	28.48 ug/m3	23.62 ug/m3	20.52 ug/m3	82.8% (PASS)
+12 Hours	31.77 ug/m3	23.63 ug/m3	21.03 ug/m3	84.6% (PASS)
+24 Hours	23.50 ug/m3	23.65 ug/m3	21.49 ug/m3	84.9% (PASS)

**Baseline Invariance & Weakest Groups: Baseline B (Seasonal-Hour) MAE is ~23.6 ug/m3 across horizons because previous 7-day occurrences for target hour (t+h) are identical for h<=24h. Weakest subgroup performance: Lahore (MAE 30.78, coverage 77.8%) and Smog Season (MAE 33.27). Post-March 2025 ground truth is an operational calibrated estimate. Karachi feed stopped transmitting in March 2025 and live forecasts are suppressed per data freshness policy.*

4. Responsible AI & Insight Agent

• **Number Guard:** Every numeric value in agent outputs is strictly verified against DuckDB Gold query results or official guidelines, preventing hallucinated numbers.
• **Data Provenance:** All forecast answers attach real database row IDs (e.g. LHE-20261020T1600Z).
• **Emergency Health Routing:** Directs acute symptoms to Punjab Emergency Rescue (1122) and Health Helpline (1033).
• **Bilingual Support:** Native Urdu templates for provincial school education administrators.
• **Unvalidated City Guard:** Multan & Faisalabad outputs are explicitly labeled with widened uncertainty bounds.